In-3-D NAND Flash Computing for Vector Similarity Search Acceleration on Edge Devices

Han-Wen Hu, *Member, IEEE*, Yuan-Hao Chang, *Fellow, IEEE*, Bo-Rong Lin, Huai-Mu Wang, Yung-Chun Lee, Hsiang-Pang Li, Chung-Kuang Chen, Tei-Wei Kuo, *Fellow, IEEE*, and Meng-Fan Chang, *Fellow, IEEE*

***Abstract*—Vector Similarity Search (VSS) on edge devices is increasingly essential for data privacy but faces substantial latency and energy overhead due to frequent data transfers between storage and DRAM. To address these challenges, we propose the Intelligent Cognition Engine (ICE), a fully digital non-volatile in-memory computing (nvIMC) framework designed for integration with commercial 3D NAND Flash. ICE avoids the use of analog-digital conversion (ADC/DAC) and reduces data movement by performing vector similarity computation directly within NAND storage. Key architectural components include a digital page multiplier, a two's-complement accumulator that supports signed computation with minimal circuit modification, and a hierarchical Top-N search strategy to reduce unnecessary data accesses. The proposed framework is evaluated through a combination of post-layout circuit simulations, representative silicon measurements, and system-level modeling on edge platforms. Experimental results indicate that ICE can achieve 17.8–122.6× speedup and 11.5–162× improvement in energy efficiency compared to conventional von Neumann-based approaches, demonstrating the feasibility and scalability of fully digital 3D NAND-based nvIMC for edge AI workloads.**

***Index Terms*—3D NAND-based nonvolatile in-memory computing (nvIMC), 3D NAND digital computing-in-memory (3D NAND dCIM), 3D NAND flash, vector similarity search (VSS).**

I. Introduction

RECENT advances in artificial intelligence (AI) have significantly increased the importance of vector similarity search (VSS) in applications such as facial recognition, content retrieval, and image classification, especially when executed locally on edge devices to ensure data security. However, conventional VSS methods on edge platforms suffer from substantial latency and energy overhead due to frequent and extensive data transfers between NAND storage and DRAM. This overhead becomes particularly pronounced in edge devices with constrained memory capacity, leading to increased search latency (tSR) and elevated search energy (E_{SR}), severely limiting their practical applicability.

Many prior architectures attempt to reduce data movement, but none fundamentally overcome this bottleneck. In-storage computing (ISC) reduces PCIe traffic yet still requires moving entire NAND pages into the controller [9], [12], [19], [21]. Analog 3D NAND nvIMC approaches exploit bitline-level current summation, but require thousands of ADC/DAC units, CSL regulators, and device-specific calibration, making them highly sensitive to V_t variation and unsuitable for TLC devices with narrow sensing windows [6], [14], [15], [16], [18], [22]. By contrast, existing digital nvIMC prototypes avoid analog non-idealities but rely on bit-serial per-BL latch operations, lack signed integer support, and do not scale to plane-parallel throughput due to tight pitch limitations. Moreover, prior digital nvIMC architectures execute TLC sensing, bitwise multiplication, and accumulation strictly in sequence, preventing any form of compute–while–sense overlap and fundamentally limiting achievable throughput. As a result, 3D-NAND-based IMC implementations expose a fundamental trade-off: *analog nvIMC is fast but unreliable under variation, whereas digital nvIMC is reliable but too slow for practical VSS acceleration.*

A key insight is that *VSS does not require high-precision arithmetic, but rather massively parallel and monotonic similarity scoring, ideally executed directly where the data reside.* Commodity TLC 3D NAND naturally offers such properties: thousands of bitlines per plane allow inherently parallel bitwise operations, and the page buffer (PB) already contains duplicated latches and logic paths originally intended for program-verify operations. These observations unlock a new design space for a fully digital compute-in-flash (dCIM) architecture in which vector–vector multiplication (VVM) is performed entirely through PB-resident digital logic—without analog current summation, ADC/DACs, or modification to BL/WL pitch or periphery circuits. During online VVM execution, ICE employs a pSLC-style read enabled by GC133 sensing, where only the effective binary state of each cell is sensed for page-buffer computation, while full TLC readout is preserved for weight storage.

However, this insight alone is insufficient to enable scalable, system-level VSS acceleration. Existing digital nvIMC

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Han-Wen Hu is with Macronix, Hsinchu 30078, Taiwan, and also with the Department of Electrical Engineering, National Tsing Hua University, Hsinchu 300044, Taiwan (e-mail: paddingtonhu@mxic.com.tw).

Yuan-Hao Chang is with the Department of Computer Science and Information Engineering, National Taiwan University, Taipei 106319, Taiwan (e-mail: johnson@csie.ntu.edu.tw).

Bo-Rong Lin, Huai-Mu Wang, Yung-Chun Lee, Hsiang-Pang Li, and Chung-Kuang Chen are with the Macronix, Hsinchu 30078, Taiwan.

Tei-Wei Kuo is with Delta Electronics and the Department of Computer Science and Information Engineering, National Taiwan University, Taipei 106319, Taiwan.

Meng-Fan Chang is with the Department of Electrical Engineering, National Tsing Hua University, Hsinchu 300044, Taiwan.

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approaches built upon latch-based in-NAND computation remain constrained by *architectural organization* rather than arithmetic precision [13]. In particular, the reliance on single-path aggregation, strictly serialized execution between TLC sensing and computation, and the absence of a system-level control and scheduling model fundamentally limit throughput scalability under realistic VSS workloads. As a result, prior digital nvIMC designs largely remain proof-of-concept demonstrations, unable to fully exploit the intrinsic plane-level parallelism of commercial 3D NAND.

This work proposes an Intelligent Cognition Engine (ICE) that addresses the fundamental scalability limitations of prior digital nvIMC designs not through incremental optimization, but by *re-architecting the digital compute fabric itself*. By enabling plane-parallel two's-complement accumulation, compute–while–sense overlap via weight-data latch decoupling, and controller-driven hierarchical search orchestration, ICE enables digital nvIMC to evolve from an isolated in-array technique toward a scalable, system-integrated substrate for real-world VSS acceleration.

To translate 3D NAND's intrinsic parallelism into practical VSS acceleration, ICE introduces two complementary throughput mechanisms that together transform latch-based digital in-NAND computation into a scalable, multi-plane compute substrate. First, a pipeline-enabled page multiplier repurposes one of the duplicated page-buffer (PB) latches as a weight-data latch (WDL), decoupling TLC sensing from bitwise BET multiplication. This organization enables compute–while–sense execution within each plane, while allowing all planes to operate concurrently. Second, ICE adopts plane-parallel two's-complement (TC) accumulation by instantiating an independent TC accumulator in each plane, reusing existing verify/counter structures already present in commercial TLC 3D NAND. This design eliminates aggregation serialization and allows all planes to execute signed BET-based accumulation in parallel, yielding 4× native compute throughput without modifying the PB layout or introducing additional arithmetic units. Together, these mechanisms form a multi-plane digital compute fabric capable of sustaining high-throughput VVM directly inside commodity 3D NAND dies.

Among these throughput mechanisms, the pipeline-enabled page multiplier is the only component that requires additional circuit resources inside the NAND die. The design introduces only two additional lightweight latches to support WDL storage and multi-bit BET operations, while preserving the original BL/WL pitch and page-buffer layout. All other compute components—including the per-plane TC accumulators—reuse existing verify/counter structures in commercial TLC 3D NAND, ensuring minimal hardware overhead and full compatibility with circuit-under-array (CuA) technology.

Beyond alleviating the throughput limitations of prior digital nvIMC architectures, ICE further incorporates several device-, architecture-, and system-level innovations that are validated through a combination of post-layout circuit simulation, representative test-chip measurements, and system-level modeling: (i) a 1-3-3 Gray code (GC133) with silicon-measured V_t window and 40M-read endurance, validating TLC sensing robustness; (ii) a formalized Bit-Error-Tolerant (BET) encoding framework for multi-precision VVM; (iii) an ICE controller architecture with plane- and die-level scheduling and pipeline orchestration, evaluated through cycle-accurate and system-level modeling; (iv) a fully modeled and evaluated Hierarchical Top-N Search (HTNS) strategy; and (v) extensive platform-level evaluation on Raspberry Pi 3/4 and Jetson Nano systems, demonstrating up to 122.6× speedup and 162× energy reduction.

This work makes the following contributions:

1) We demonstrate a scalable, system-integrated digital in-NAND computation framework for vector similarity search on commercial TLC 3D NAND.
2) We introduce architectural mechanisms—including plane-parallel two's-complement accumulation and compute–while–sense pipelining—that remove the serialization bottlenecks inherent in prior digital nvIMC designs.
3) We develop a unified error-resilient design spanning TLC sensing (GC133), data representation (BET), and system orchestration (HTNS), enabling reliable VVM without invoking ECC on the critical path.
4) We provide a comprehensive evaluation that spans circuit-level analysis, representative silicon measurements, and system-level modeling based on real edge platforms.

II. Background, Limitations & Motivation

A. Vector Similarity Search (VSS) Study

1) Background on VSS: This section provides an overview of the Vector Search and Similarity (VSS) system in AI search applications.

General AI search applications typically use a Siamese feature representation network architecture [8], [10]. This network, during its training phase, learns to generate similarity scores by comparing a query input with dataset feature vectors. After training, the model facilitates VSS in two phases: offline and online, as depicted in Fig. 1.

In the offline phase, each dataset item is processed through machine learning methods to create a feature vector (❶), stored in a database (❷). The online phase involves receiving a query in the form of a feature vector (❸), followed by computing similarity scores between this query and the database feature vectors (❹).

The system then ranks these scores to identify the top-k patterns closely matching the query (❺) and outputs relevant information based on these top-k results.

Considering a query input $x \in \mathbb{R}^d$ and a feature vector dataset $[w_i]_{i=0:n}$ $\left(w_i \in \mathbb{R}^d\right)$, the top-k similarity search result L using $L2$ distance (Euclidean distance) is:

$$L = k - argmin_{i=0:n} \|x - w_i\|_2 \tag{1}$$

Here, d represents the vector dimension, and n the number of feature vectors. We can decompose the equation as:

$$\|x - w_i\|_2^2 = \|x\|^2 + \|w_i\|^2 - 2\,(x \cdot w_i) \tag{2}$$

The first two terms can be precomputed in a single iteration over x and matrix w. The computational challenge in VSS is the final term $((x \cdot w_i))$, a vector-matrix multiplication, further reducible to vector-vector multiplications (VVM).

2) VSS System-Level Perspective: Vector Similarity Search (VSS) has become a core component in modern edge-AI applications such as face recognition, multimodal authentication, content retrieval, and personalized services. A Siamese network generates high-dimensional embeddings during the offline stage, while the online stage computes similarity scores between an input vector and a large stored dataset (Fig. 2).

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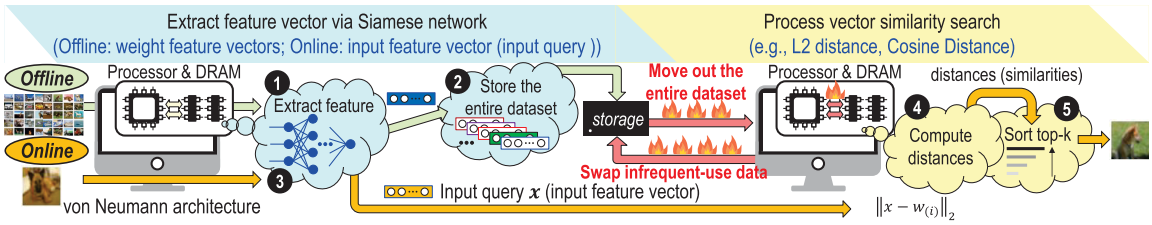


Fig. 1. The general workflow of VSS system with Siamese feature representation network.

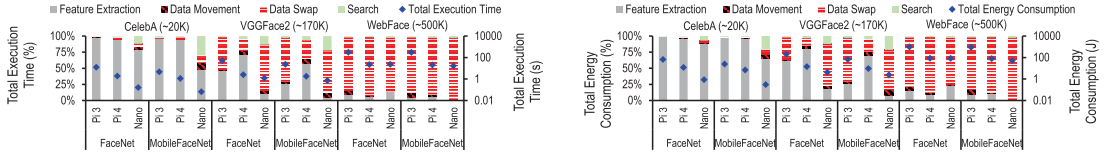


Fig. 2. Measured system performance for implementing traditional face search on various edge devices. The searching time with large datasets (e.g., CASIA-WebFace, a 500K-face dataset) will dominate the total execution time on edge devices.

Although conceptually simple, VSS exposes fundamental challenges when executed on resource-constrained edge devices.

VSS latency on edge devices is dominated by data movement rather than computation. Vector Similarity Search (VSS) on edge devices is fundamentally constrained not by arithmetic throughput, but by how quickly data can be moved through the memory hierarchy. Our measurements on Raspberry Pi3 [4], Raspberry Pi4 [5], and NVIDIA Jetson Nano [2] using datasets of 20K (CelebA [17]), 170K (VGGFace2 [7]), and 500K (CASIA-WebFace [23]) feature vectors reveal a consistent upward shift in bottlenecks: feature extraction dominates for small datasets, similarity scoring dominates for medium datasets, and ultimately, data movement and DRAM swapping dominate for large datasets (Fig. 2). Across all platforms, 90–99% of execution time and energy stems from repeatedly transferring feature vectors between NAND and DRAM, rather than performing vector arithmetic. This reflects a fundamental limitation of the memory hierarchy: VSS throughput is governed by data bandwidth, not multiplier performance.

Limited DRAM capacity amplifies memory-hierarchy imbalance. This data-movement bottleneck is further worsened by the tight DRAM budgets of edge devices. As dataset size grows, feature vectors cannot remain fully resident in DRAM, causing the operating system to swap memory pages frequently. Once swapping begins, DRAM bandwidth becomes saturated, and even relatively powerful processors stall waiting for data. As long as tens or hundreds of megabytes of feature vectors must be fetched repeatedly from NAND storage, simply increasing CPU/GPU compute capacity cannot meaningfully improve VSS performance. The system becomes inherently bandwidth-bound, not compute-bound.

Existing solutions fail because none eliminate the root cause: data movement. A variety of acceleration strategies have been proposed, yet all of them inherit the same fundamental limitation. Compute accelerators (CPUs, GPUs) still depend on DRAM and therefore remain constrained by storage to DRAM traffic. ANN and indexing techniques may prune arithmetic operations, but they do not reduce the amount of vector data that must be loaded into DRAM. In-storage computing (ISC) removes PCIe transfer overhead but still requires complete NAND pages to be delivered to the controller. Analog 3D-NAND nvIMC offers high theoretical throughput, but depends on CSL current summation, multi-range sensing, calibration loops, and thousands of ADCs—rendering it highly sensitive to V_t variation and impractical for TLC Flash. Prior digital nvIMC prototypes avoid analog instability but suffer from bit-serial PB execution, lack signed arithmetic support, and are unable to pipeline or exploit plane-level parallelism, resulting in insufficient throughput for real VSS workloads.

System-level requirements for a practical VSS accelerator. These observations lead to a single conclusion: none of the existing solutions remove the true source of VSS latency and energy—massive, repeated NAND to DRAM data movement required for VVM operations. A practical, system-level solution needs to satisfy four requirements simultaneously: (1) perform VVM directly inside NAND, removing data transfers entirely; (2) exploit the native parallelism of 3D NAND at the bitline, plane, and die levels; (3) operate reliably on TLC NAND, to keep the NAND capacity and tolerate bit errors; and (4) avoid analog circuitry, ensuring manufacturability, robustness, and compatibility with commercial CuA NAND. These system-level demands motivate the development of a digital dCIM architecture that integrates multi-plane scalability, compute–sense pipelining, TLC-friendly sensing, and error-tolerant encoding. The proposed ICE framework, described in Section III, is designed precisely to address these challenges and convert 3D NAND's intrinsic storage parallelism into practical VSS acceleration.

B. 3D NAND-Based nvIMC Limitations

3D NAND Flash stores data in vertically stacked memory cells, each accessible through thousands of parallel bitlines (BLs). During read operations, BL currents are sensed simultaneously through page-buffer (PB) circuits (Fig. 3). This naturally provides massive fine-grained parallelism, motivating near-data processing such as in-memory VVM. Over the years, both analog nvIMC and digital nvIMC approaches have been proposed to unlock this parallelism. However, applying these approaches to commercial TLC 3D NAND reveals fundamental scaling limitations. Our experiments with 96-layer 3D NAND, using CuA technology, have uncovered several unaddressed design challenges in analog and digital nvIMC implementation on 3D NAND. These challenges are detailed in the ensuing paragraphs.

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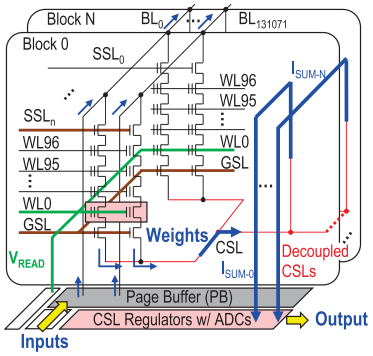


Fig. 3. The concept of our 3D NAND-based analog nvIMC implementation using 96-layer 3D NAND with the circuit under array (CUA) technology.

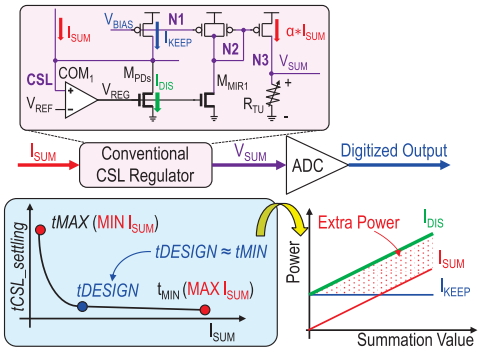


Fig. 4. Design challenge-2: wide range summation current of using the conventional CSL regulator.

1) Analog 3D NAND*-Based nvIMC Limitations:* **Analog nvIMC faces intrinsic obstacles imposed by TLC sensing and CSL summation.** Analog nvIMC [6], [14], [15], [16], [18], [22] performs computation by generating per-cell currents and summing them along the CSL or BLs before digitization by ADCs. Although TLC provides eight threshold-voltage levels, analog nvIMC cannot safely utilize them for computation. Instead, each TLC cell is effectively reduced to an SLC-like 0/1 current source, because the absolute cell current is highly sensitive to V_t variation, read disturb, temperature drift, and P/E cycling. These device-level effects distort the accumulated current I_{SUM}, preventing it from representing the ideal linear sum required for VVM.

CSL summation currents vary widely, forcing expensive current regulation. The magnitude of I_{SUM} depends heavily on input data patterns, producing a very wide dynamic range of summation currents. As shown in Fig. 4, the CSL regulator must stabilize the CSL voltage across this wide range, which requires large I_{KEEP} bias currents and results in long settling times. These requirements substantially increase both power consumption and compute latency, making analog accumulation increasingly impractical as VVM dimensionality grows.

High-dimensional VVM requires thousands of ADCs, leading to prohibitive area overhead. Even if CSL variation could be controlled, analog nvIMC still faces scaling limitations due to its reliance on ADCs. High-resolution, high-speed ADCs are required for every BL or every partial-sum path. For a 512-dimensional INT8 VVM, prior works require 2,048 to 8,192 ADCs, which translates into >4,000% area overhead—far beyond what commercial TLC NAND can accommodate. This alone makes analog nvIMC unsuitable for manufacturable products.

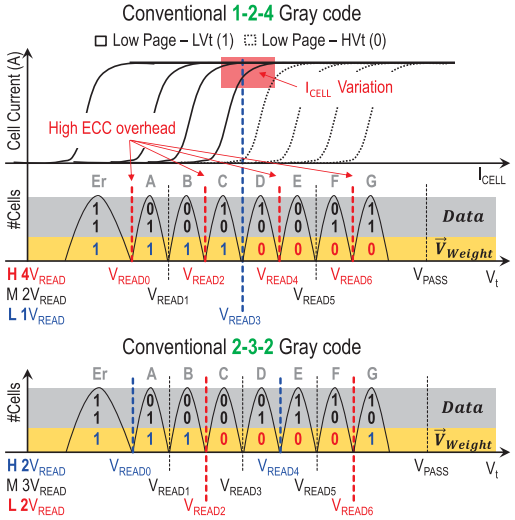


Fig. 5. Design challenge-1: using 2-3-3 Gray code needs multiples V_{READ} to distinguish low virtual page, and using 1-2-4 Gray code faces ECC issue on high virtual page.

Another inherent limitation is that analog nvIMC cannot naturally represent negative values, as this requires bidirectional current generation. Such circuitry is incompatible with the compact NAND periphery and CSL routing. This prevents analog nvIMC from supporting signed VVM, which is required for INT4/INT8 similarity scoring.

Due to unstable current summation, expensive CSL regulation, and massive ADC requirements, analog nvIMC—even though promising in concept—cannot be manufactured in commercial TLC NAND and cannot scale to real-world VSS workloads. These limitations motivate a transition toward a digital nvIMC architecture.

2) Digital 3D NAND*-Based nvIMC Limitations:* **Digital nvIMC avoids analog variability but is limited by TLC multi-level sensing latency.** Unlike analog designs, digital nvIMC uses PB latch operations rather than cell current, making it inherently immune to analog variation and compatible with dense TLC storage. However, TLC's multi-level nature introduces a different bottleneck for dCIM: conventional Gray-code sensing requires multiple V_{READ} cycles to distinguish middle/high virtual pages. Codes such as 1-2-4 and 2-3-2 (Fig. 5) and amplify ECC demand at high-page boundaries. As a result, sensing latency—not computation—becomes the limiter for digital nvIMC throughput. If not addressed, multi-V_{READ} sensing prevents dCIM from sustaining VVM pipelines and restricts the attainable system-level speedup.

Existing digital nvIMC also lacks architectural mechanisms to fully exploit NAND parallelism. Although 3D NAND inherently provides massive structural parallelism across thousands of bitlines and multiple planes, neither analog nor existing digital nvIMC architectures are able to translate this parallelism into usable computation throughput. Prior design [13] employ a single shared accumulator per die, forcing all PB-generated partial products to serialize through one aggregation path. Without the ability to perform accumulation independently on each plane and without a means to overlap PB logic with ongoing TLC sensing, prior digital nvIMC architectures cannot capitalize on the multi-plane structure of 3D NAND. Furthermore, existing digital nvIMC designs serialize the entire VVM flow: TLC sensing must complete before any PB-level multiplication can begin, and accumulation cannot

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proceed until all bitwise operations finish. This rigid sequencing forces sensing, multiplication, and accumulation to occur strictly one after another, eliminating any opportunity for compute–while–sense pipelining. As a result, the attainable throughput is fundamentally capped by the slowest stage, leaving most of NAND's intrinsic parallelism unused. This architectural bottleneck—not device limitations—ultimately caps the attainable throughput and leaves most of NAND's intrinsic parallelism unused.

These limitations highlight a fundamental gap between NAND's potential parallelism and the practicality of existing nvIMC approaches. Analog nvIMC is constrained by unstable SLC-like current summation, wide-range CSL regulation, excessive ADC overhead, and the inability to represent negative weights—making it incompatible with TLC-based systems. Digital nvIMC, while robust against analog variation, remains bottlenecked by multi-level TLC sensing latency and the strictly bit-serial nature of PB operations, which prevents it from scaling with the available bitline parallelism. To bridge this gap, an nvIMC framework must *rethink how computation and sensing can interact inside* NAND *so that the architecture operates reliably on TLC devices while sustaining high computational throughput.*

C. Motivation

Digital nvIMC is required to avoid the fundamental overhead and instability of analog computation. As outlined in the previous section, analog nvIMC faces intrinsic obstacles that prevent its deployment in commercial TLC NAND: unstable SLC-like current summation, wide-range CSL regulation, excessive ADC requirements, and lack of signed arithmetic. These are not engineering inconveniences but structural limitations of current-summing architectures in multi-level Flash. A practical in-flash compute solution must therefore be fully digital, relying exclusively on latch-based operations that are robust to V_t variation, compatible with TLC memory structures, and manufacturable within the strict pitch constraints of commercial arrays.

Reliable VMM requires error resilience embedded directly into sensing and data representation—not delegated to ECC. Digital nvIMC must operate without invoking ECC, as ECC decoding disrupts the VVM pipeline and introduces nondeterministic latency. Yet TLC readout inherently suffers from boundary noise, V_t variation, and read-disturb effects. Ensuring reliable in-NAND computation therefore requires a unified resilience strategy in which sensing robustness and data-level redundancy complement each other: the sensing process must remain stable without increasing ECC pressure, and the stored vectors must incorporate intrinsic redundancy so that bit errors minimally perturb similarity scores. These requirements motivate the sensing- and encoding-oriented design directions adopted in ICE.

Signed VMM requires a representation compatible with NAND's pitch constraints. Similarity search requires signed arithmetic, yet the NAND periphery cannot accommodate subtractors or adders. A practical dCIM architecture must therefore support sign-aware accumulation using only lightweight latch-based operations, motivating the representational and arithmetic principles employed in ICE.

Throughput scalability requires architectural, not device-level, changes. A key obstacle is that prior digital nvIMC architectures are inherently serialized: TLC readout, bitwise multiplication, and accumulation occur strictly one after another, preventing overlap between these stages. Such serialization stalls the compute pipeline during every sensing cycle and restricts throughput regardless of available BL or plane parallelism. Even with robust sensing and data encoding, digital nvIMC cannot provide meaningful VSS acceleration unless it exploits the massive parallelism already present in 3D NAND. This requires three architectural capabilities: (1) compute-while-sense pipelining, so that PB logic can operate on previously sensed data while the array performs new reads; (2) plane-parallel accumulation, enabling each plane to independently aggregate partial sums rather than serializing all operations through a single accumulator; and (3) search-space reduction, so that only the necessary weight vectors are accessed during VSS. These architectural enablers are essential for converting NAND's structural parallelism into sustained end-to-end throughput.

ICE therefore integrates robust TLC sensing, redundancy-aware encoding, compute–while–sense pipelining, plane-parallel accumulation, and hierarchical Top-N filtering to form a manufacturable and high-throughput digital compute-in-flash architecture. Section III elaborates the mechanisms enabling these capabilities.

III. Proposed 3D NAND-Based VSS Solution: ICE

ICE is designed to satisfy the key requirements identified in Section II: (1) a fully-digital computation path manufacturable in TLC NAND; (2) built-in error resilience at both the sensing and data-representation levels; (3) architectural mechanisms for high-throughput compute–sense pipelining and plane-parallel execution; and (4) system-level pruning of unnecessary vector evaluations. This section presents the architectural and circuit components that collectively realize these capabilities.

A. ICE Architecture

Fig. 6 illustrates the ICE architecture, which integrates four cognitive TLC 3D NAND chips with a lightweight controller that coordinates sensing, computation, and hierarchical filtering. The architecture consists of five tightly coupled subsystems, each corresponding to a major block in the figure.

1) **System–level front end (HTNS).** The controller first performs Hierarchical Top-N Search (HTNS) module to prune irrelevant partitions (Fig. 7) before triggering fine-grain in-NAND VVM, reducing unnecessary NAND accesses and improving end-to-end latency. As shown on the left side of Fig. 6, HTNS reduces the number of NAND pages that participate in fine-search, serving as the entry point of the ICE workflow. (Fig. 7)
2) **Query input path (L2–L5 binary expansion).** The query embedding is represented in its native 4-bit (or 8-bit) form, without error-tolerant encoding, since it does not traverse the noisy TLC channel. Its individual bits are loaded into PB latches L2–L5, where they serve as binary selectors for the corresponding unary lanes used during BET-based multiplication.
3) **TLC weight sensing layer (GC133).** Stored weight vectors are retrieved from TLC cells through GC133 sensing and latched into the weight-data latch (WDL). Because the weights originate from noisy TLC storage, this path carries the error-prone components of the computation. This sensing operation—shown in the sensing block of Fig. 6—preserves single-V_{READ} low-page access while improving robustness for upper pages.

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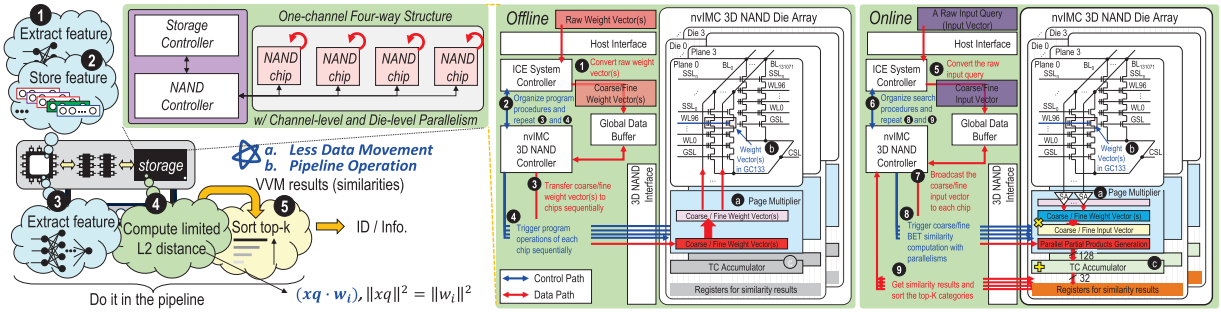


Fig. 6. Conceptual architecture and workflow of the proposed 3D NAND-based VSS system architecture, i.e. intelligent cognitive engine (ICE), and the relative workflow. The edge processor serves as a feature extractor, while ICE performs in-NAND VVM and hierarchical search to reduce data movement. ICE can contribute to the offline phase of storing different datasets and the online phase of searching for the most similar answer to suppress return data. 3D NAND dies are with nvIMC techniques, featuring the purely digital nvIMC designs with (a) a 1-3-3 Gray code to scramble TLC cells for data storing and add a buffer zone in V_t window to tolerate error bits, (b) four digital page multipliers for parallel bitwise logic operations, and (c) four two's complement (TC) accumulators to support signed computations. Compute–while–sense overlap occurs within each plane via WDL decoupling, while multiple planes execute concurrently.

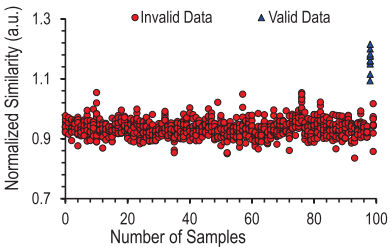


Fig. 7. Valid data vs invalid data in a partial dataset of VGGFace2 (8000 photos w/ 100 people).

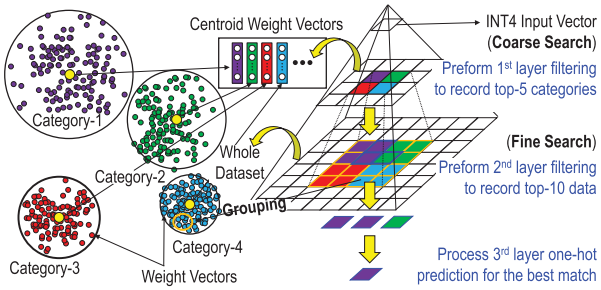


Fig. 8. The concept of Hierarchical Top-N Search (HTNS) scheme. HTNS aims to largely decrease unnecessary data computations during search and provide the same search accuracy compared to brute-force search.

4) **In-NAND compute layer (page multiplier + per-plane TC accumulator).** Once WDL—realized by reusing the existing L1 latch—is loaded, the page-buffer logic performs bitwise multiplications between the sensed weight bits and the binary query bits stored in L2–L5. These operations activate multiple unary-weight lanes in parallel, producing several partial-product streams that feed into the per-plane TC accumulators.
5) **Parallelism layer (multi-dimensional pipelining).** ICE exploits three forms of parallelism—(i) intra-plane compute–sense overlap, where GC133 sensing and PB-level BET VVM proceed concurrently within the same plane, enabled by the weight-data latch (WDL), and (ii) plane-parallel accumulation—jointly reflected in the overlapping activation of planes in Fig. 6. This layer overcomes the strict serialization of prior digital nvIMC.

Together, these subsystems compose the static structure of ICE. Section III-B describes how they operate dynamically during a complete VSS query.

B. End-to-End Workflow of ICE

The execution sequence in Fig. 6 shows how coarse filtering, TLC sensing, and in-NAND computation cooperate to realize digital VVM.

1) Coarse-Stage Filtering via HTNS: The controller receives the query embedding and applies HTNS to identify the most relevant partitions, reducing unnecessary NAND accesses (Fig. 8).

2) Query Bit Loading Into PB Latches (L2–L5): The query's 4-bit (or 8-bit) components are directly expanded into PB latches L2–L5. Since the query does not pass through TLC storage, no BET redundancy is applied; the PB simply holds the native binary values that determine which unary lanes participate in multiplication.

3) Compute–While–Sensing With Intra-Plane Overlap: Within each activated plane, GC133 sensing and PB-level BET VVM are overlapped by decoupling the sensed weight bits through WDL. All planes selected by HTNS execute this overlapped sensing and computation concurrently, without time-multiplexed execution across planes. Steps are as follows:

a) TLC Weight Sensing Using GC133: For each activated partition, the TLC wordline storing the weight vector undergoes GC133 sensing. The sensed weight bits are latched into WDL. This path constitutes the error-prone component of VMM.

b) Unary Lane Activation and Bitwise Multiplication: The PB evaluates the unary-weight lanes selected by the query bits in L2–L5, producing parallel partial products.

c) Per-Plane TC Accumulation: Each plane independently aggregates unary partial sums using a duplicated TC accumulator, producing one signed VVM result per plane, as implemented by the per-plane TC accumulators shown in Fig. 12.

In ICE, the controller broadcasts commands to all planes selected by HTNS. Each plane independently overlaps GC133 sensing and PB-level BET VVM via WDL, while the controller only coordinates high-level activation and result collection, without inter-plane time-multiplexed scheduling.

4) Final Top-N Selection: The ICE controller merges results across planes and dies and outputs the Top-N matches.

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TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT GRAY CODES

Sensing Scheme	TLC Read for storage/ pSLC-style Read for dCIM Characteristics			
	V_{READ} Cycles (Low, High)	Target RBER (Typical)	tR_TLC , tR_pSLC	ECC Dependency
1–2–4 Gray Code	1, 4	High (High Page)	92, 50 (us)	High
2–3–2 Gray Code	2, 2	Low	77, 64 (us)	Moderate
GC133 (This work)	**1, 3**	**Low**	**77, 50 (us)**	**Moderate**

ECC dependency indicates whether ECC must be invoked during VVM execution to maintain acceptable accuracy.

Through this workflow, ICE jointly exploits HTNS-based search-space reduction, GC133 sensing robustness, BET error-tolerant representation, PB-level parallel multiplication, and plane-level TC accumulation. The result is a fully digital, multi-dimensional pipelined compute-in-flash engine that achieves high throughput while preserving full compatibility with commercial TLC NAND.

C. GC133 TLC Sensing for Reliable Digital Readout

Digital nvIMC eliminates the analog current-summation variations inherent to prior CIM designs, but its reliability ultimately depends on the correctness of the TLC readout. Unfortunately, multi-level TLC sensing suffers from narrow V_t margins, asymmetric state drift, and boundary overlap, which directly translate into bit errors inside the weight-data latch (WDL). Conventional TLC Gray codes are inadequate in this context. The widely used 1–2–4 code preserves single-V_{READ} low-page access but offers limited margin for middle and high pages, which increases error rates when ECC is not invoked during VVM. Conversely, the 2–3–2 code expands the margin for upper pages but requires additional V_{READ} cycles, stalling the compute pipeline and degrading throughput.

To overcome these limitations, ICE adopts the proposed **1–3–3 Gray code (GC133)**, designed specifically for digital in-flash computation where both throughput and sensing robustness are essential. GC133 retains **single-V_{READ} low-page discrimination**, enabling sensing to supply PB computation without stalling. For the middle and high pages, GC133 introduces **three V_{READ} thresholds** per page, which widens the effective V_t margin and reduces boundary misclassification. For middle and high pages, GC133 still employs multi-threshold TLC sensing to preserve storage density, while pSLC-style read is selectively used only to supply binary weight bits during online VVM execution. Measurements from a 96-layer TLC test chip (Fig. 9) confirm that GC133 increases the noise tolerance of mid/high pages while preserving fast low-page performance.

Once GC133 sensing is completed, the decoded weight bits are latched into **WDL** in binary form. Although WDL itself stores no redundancy, the BET-expanded pattern programmed in the TLC array ensures that the resulting weight bits remain resilient to occasional read errors. GC133 thus provides the device-level robustness needed to feed reliable weight bits into the subsequent BET-based digital VVM engine. Table I summarizes the quantitative trade-offs among conventional TLC Gray codes and the proposed GC133 in terms of read latency, error robustness, and ECC dependency.

D. Error-Resilient Weight Encoding for Digital VVM

Reliable digital VVM inside TLC NAND requires tolerance to raw bit errors without invoking ECC during computation, since ECC decoding disrupts pipeline execution and introduces nondeterministic latency. While GC133 improves the robustness of TLC sensing, the stored weight representation itself must further mitigate errors caused by retention loss, read disturb, and V_t variation. ICE therefore adopts a *Bit-Error-Tolerant (BET)* encoding scheme that applies *only to stored weight vectors*. The query input does not traverse the noisy TLC channel and thus remains in its native binary form, with individual bits directly loaded into page-buffer latches (L2–L5).

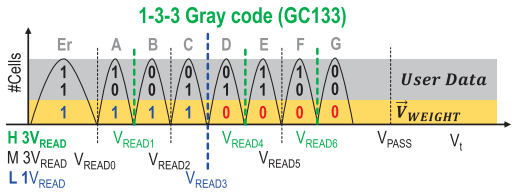


Fig. 9. The proposed 1-3-3 Gray code (GC133) only uses single V_{READ} to distinguish low virtual page and is with lower ECC issue compared to two conventional Gray codes.

1) Binary Weight Representation: Let a signed K-bit weight be represented as

$$w = s \cdot \sum_{k=0}^{K-1} b_k \cdot 2^k, \tag{3}$$

where $s \in \{+1, -1\}$ denotes the sign and $b_k \in \{0, 1\}$ are the binary magnitude bits.

2) General BET Formulation: In BET, each binary bit b_k is mapped to a group of unary replicas:

$$\mathrm{BET}(w) = \left\{ s, \bigcup_{k=0}^{K-1} \{ u_{k,1}, u_{k,2}, \ldots, u_{k,c_k} \} \right\}, \qquad u_{k,i} = b_k, \tag{4}$$

where c_k denotes the replication factor associated with the significance level 2^k. Unlike full unary expansion, BET employs fixed, small replication factors to balance error resilience and storage overhead.

3) Error-Tolerance Property: With BET, the contribution of each significance level can be expressed as

$$w = s \cdot \sum_{k=0}^{K-1} \left(\frac{1}{c_k} \sum_{i=1}^{c_k} u_{k,i} \right) \cdot 2^k. \tag{5}$$

As a result, a single unary-bit error at level k introduces a bounded error

$$\Delta w_k = \pm \frac{2^k}{c_k}, \tag{6}$$

which is significantly smaller than the $\pm 2^k$ error caused by a binary MSB flip. By assigning larger c_k to more significant bits, BET substantially suppresses the impact of MSB errors, which are the dominant source of accuracy degradation in TLC-based storage.

4) Implementation-Aware Bet Configuration in ICE: In practice, ICE adopts a compact BET configuration with fixed small replication factors to minimize overhead. For INT4 weights ($K = 4$), ICE uses $\{c_k\} = \{1, 1, 2, 2\}$ which expands each 4-bit signed weight into six unary weight bits. Fig. 10 provides a concrete illustration of this 6-bit BET expansion and the corresponding unary-weight grouping for INT4 VVM. This design significantly reduces MSB error sensitivity while incurring only modest storage overhead, in contrast to full unary expansion.

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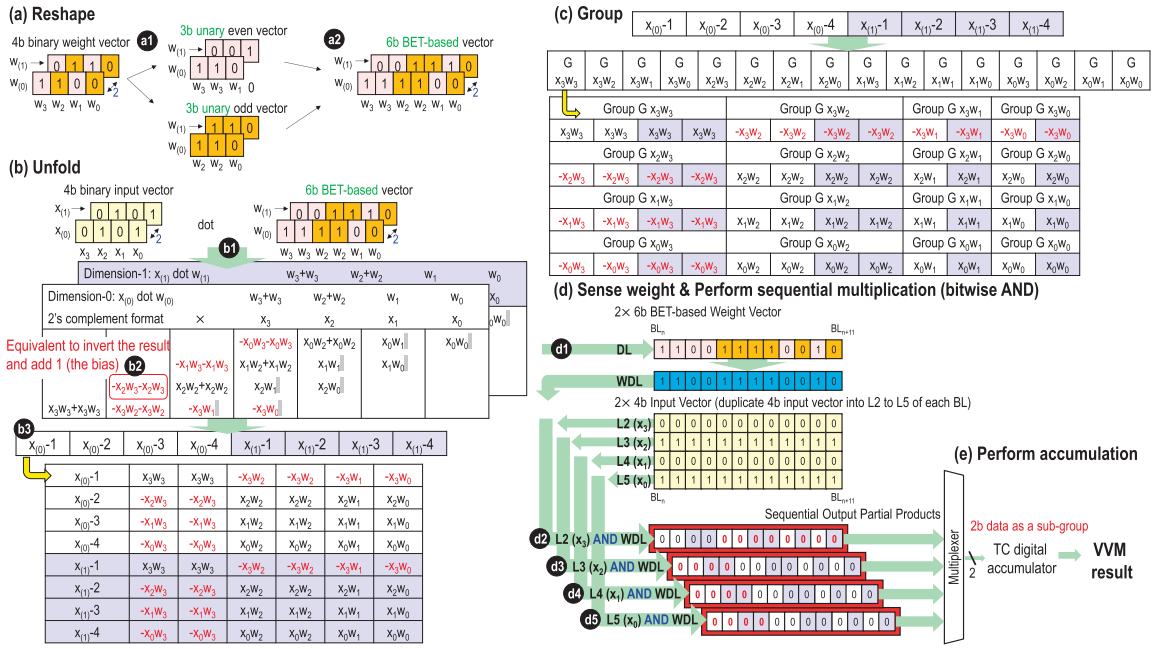


Fig. 10. Bit-error-tolerance (BET) VVM flow. To achieve BET VVM, it needs five major steps. (a) Reshape the weight vector. (b) Unfold the multiplications. (c) Group the same weighted value. Then, (d) sense weight and perform sequential bit multiplication, and (e) bit accumulation to get the VVM result. Note that this is a short example for illustration, and ICE could support higher dimension and data precision vectors (e.g., 512-dimensional 4b vectors).

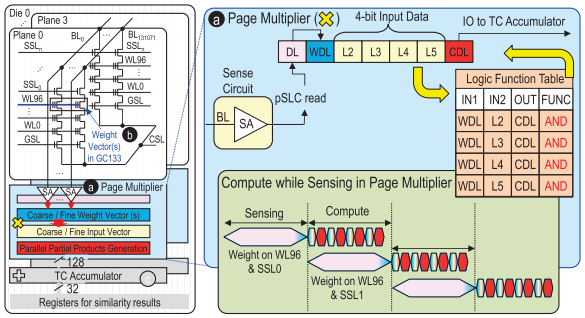


Fig. 11. The page multiplier operates by combining (1) pSLC-style read using GC133 sensing with (2) bitwise AND operations implemented via the program-verify logic function table for BET-based VVM. The design reuses the existing L1 latch as the weight-data latch (WDL) and introduces only two additional latches (L4 and L5) to support concurrent query expansion, enabling compute–while–sense execution within the page buffer.

5) Data-Path Realization: The unary replicas defined by BET are programmed into the TLC array as part of the weight storage pattern. During computation, GC133 sensing decodes these cells into binary form and latches the results into the weight-data latch (WDL). d1 Although redundancy is not explicitly visible in WDL, it remains functionally effective through the underlying TLC storage pattern. The binary query bits stored in L2–L5 act as unary-lane selectors that activate the corresponding BET weight replicas. The page buffer then performs parallel bitwise AND operations (Fig. 10 d2 to d5, and the resulting unary-weighted partial products are aggregated by the per-plane two's-complement (TC) accumulators to produce signed VVM results (Fig. 10 e. This unary-weighted multiplication and sign handling process is exemplified in Fig. 12, which serves as a step-by-step illustration of the BET-based VVM procedure.

By applying BET exclusively to stored weights and preserving a binary query path, ICE avoids unnecessary encoding overhead while achieving strong resilience to TLC-induced errors. Combined with GC133 sensing and multi-dimensional pipelining, BET forms the data-layer foundation that enables reliable, high-throughput, and fully digital VVM in commodity TLC 3D NAND.

E. Digital Compute Fabric

ICE transforms the TLC NAND page-buffer array into a fully in-NAND digital compute fabric that eliminates the strict serialization of prior digital nvIMC architectures. Unlike earlier designs, which execute TLC sensing, bitwise multiplication, and accumulation sequentially, ICE enables these stages to proceed in parallel through architectural decoupling and multi-plane execution.

Pipeline-enabled page multiplier for unary lane activation and bitwise multiplication. Commercial TLC NAND page buffers typically provide a limited set of latches, including the data latch (DL), L1, L2, L3, and the cache data latch (CDL). In ICE, the existing L1 latch—which is unused in standard read operations—is repurposed as the *weight-data latch* (WDL) to store the binary weight bits decoded from pSLC-style read using GC133 sensing for VVM execution. To support compute–while–sense execution and multi-bit BET operations, ICE introduces only two additional lightweight latches (L4 and L5), while preserving the original BL/WL pitch and page-buffer layout. This organization decouples page-buffer computation from ongoing TLC readout and enables concurrent query expansion and bitwise logic evaluation.

The query embedding remains in native binary form and is directly loaded into PB latches L2–L5, which act as *binary selectors* to determine the active unary replicas of the BET-encoded weights. For each bitline, the page buffer reuses the program-verify logic function table to perform parallel bitwise AND operations between the binary weight bits in WDL and the binary query bits in L2–L5. Meanwhile, the NAND array senses the next weight wordline and latches the

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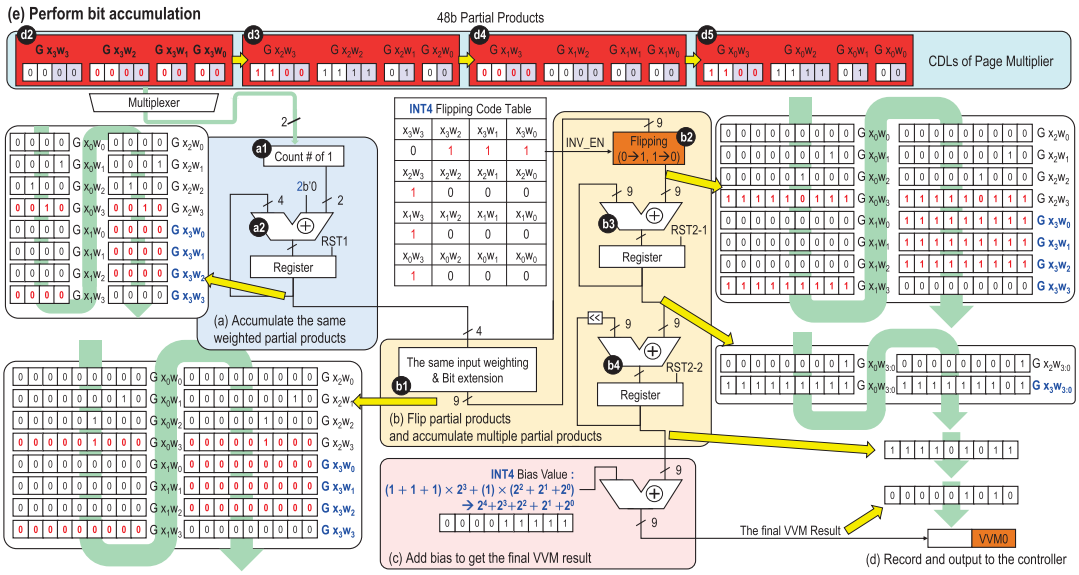


Fig. 12. Per-plane Two's-Complement Accumulator, and the registers for buffering the final similarity results.

results into the data latch (DL), allowing page-buffer computation and array sensing to proceed concurrently and enabling compute–while–sense execution in the page multiplier.

Per-plane two's-complement accumulation.

As illustrated in Fig. 12, ICE equips each plane with an independent two's-complement (TC) accumulator, constructed by reusing the verify/counter and latch structures already available in commercial TLC NAND. Rather than introducing arithmetic units, the TC accumulator operates on unary-weighted partial products generated by the page buffer.

Specifically, positive and negative contributions are handled through sign-aware control of the accumulation path. As shown in Fig. 12, the sign bit of the stored weight determines whether the corresponding unary partial products are accumulated directly or first inverted and biased before accumulation. This mechanism realizes two's-complement behavior using only latch inversion and bias addition, avoiding subtractors that cannot be supported under tight BL/WL pitch constraints.

Because the TC accumulator is instantiated independently in each plane, unary partial products are aggregated locally without serialization across planes. This design removes the single-accumulator bottleneck of prior digital nvIMC prototypes and enables four parallel signed VVM pipelines per die, as explicitly depicted in Fig. 12.

Multi-dimensional pipelining. By combining WDL-based decoupling with per-plane accumulation, ICE executes VVM using three complementary forms of parallelism: (i) intra-plane compute–sense overlap, where pSLC-style readout of AND and accumulation, TLC-stored weights and PB-level BET VVM proceed concurrently within each plane; (ii) intra-plane unary parallelism, where a single TLC-stored weight page, sensed in pSLC-style binary form, activates multiple BET unary replicas through the binary selectors in L2–L5; and (iii) plane-parallel accumulation, where all activated planes aggregate partial sums concurrently using independent TC accumulators.

F. Data Management in Cognitive 3D NAND

Role of ECC and its impact on the critical path.

Although TLC NAND inherently requires error correction to ensure long-term data integrity across retention and endurance

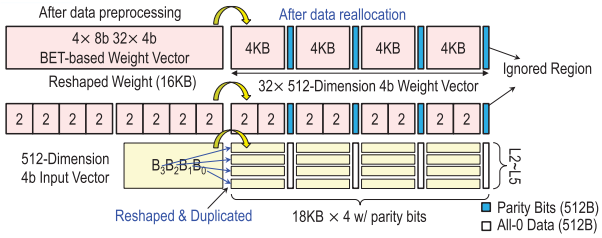


Fig. 13. Data mapping of the physical page for parity bits and all-0 data. The partial products of these regions would be zero, maintaining the VVM results without influence.

cycles, it is important to clarify that ECC decoding is not on the critical path of VVM execution in ICE. In the proposed architecture, ECC is invoked only during background maintenance operations such as data refresh, reprogramming, or integrity verification, and is never exercised during online BET-based VVM computation.

During VVM execution, ICE operates directly on raw TLC readouts produced by GC133 sensing and BET-encoded weight representations. To avoid interference from parity information, bitlines corresponding to ECC parity are explicitly excluded from accumulation, as illustrated in Fig. 13. As a result, ECC decoding neither introduces latency nor consumes energy on the VVM critical path, preserving deterministic execution timing and enabling full compute–while–sense pipelining. This design choice is essential for system-level scalability. Invoking ECC during every VVM operation would stall the pipeline and introduce non-deterministic latency, negating the benefits of in-NAND computation. By decoupling ECC from online computation and confining it to infrequent background operations, ICE achieves reliable long-term storage while maintaining high-throughput, low-energy VVM execution.

1) Data Precision: There is a tradeoff among similarity search speed, accuracy, and data precision in ICE. By adopting the BET data encoding and the data layout shown in Fig. 13, a BET-based vector with 8b precision occupies 32,768 ($2^4 \times 2 \times 512 \times 2$) bits for 8b weight value, 512-dimension feature vectors, and 2 groups for 8b input value. If we reduce the precision to 4b, the size of each vector reduces to 4,096

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$(2^2 \times 2 \times 512)$ bits, allowing us to support 8× the number of vectors in a single page. That is, ICE's performance can be further enhanced by leveraging low-precision inputs/weights according to the accuracy needed for different VSS applications.

IV. Experiment Setup

A. Workload

We evaluate ICE on face search tasks using CelebA (20K images) [17], VGGFace2 (170K) [7], and CASIA-WebFace (500K) [23]. Feature vectors are extracted via pretrained networks, normalized, and quantized to 512-D INT4 for BET VVM. The original feature vector dataset in VGGFace2 (resp. CelebA, CASIA-WebFace) requires 340MB (resp. 40MB, 1GB) of memory capacity with 170K (resp. 20K, 500K) faces of 500 (resp. 1,000, 10,575) individuals. After adopting HTNS, the coarse search dataset in VGGFace2 (resp. CelebA, CASIA-WebFace) requires 192KB (resp. 384KB, 4.1MB) of memory capacity with 500 (resp. 1,000, 10,575) $\vec{V}_{WEIGHT_C}$ vectors. The 4-bit fine search dataset of VGGFace2 (resp. CelebA, CASIA-WebFace) requires 65.3MB (resp. 7.68MB, 192MB) of memory capacity with 170,000 (resp. 20,000, 500,000) $\vec{V}_{WEIGHT_F}$ vectors. Each $\vec{V}_{WEIGHT_C}$, and $\vec{V}_{WEIGHT_F}$ are all 512-dimension INT4 vectors. HTNS reduces fine-search dataset size dramatically (e.g., VGGFace2: 340 MB to 192+653 KB, CelebA: 40 MB to 384+38.4 KB and, CASIA-WebFace: 1GB to 4100+90 KB), lowering in-memory data requirements per search and enabling efficient searches in resource-constrained devices.

B. Systems for Benchmark

For the device-level evaluation of similarity computation, we use HSPICE [11] for post-layout simulation to measure the energy consumption of modified circuits and performance. All reported circuit simulations are performed at the TT process corner, nominal supply voltage, and room temperature (25 °C). We further implement a nvIMC 3D NAND test chip on 96-layer 3D NAND with the CuA technology. The nvIMC 3D NAND test chip can demonstrate the validations of the proposed functions. For the system-level evaluation, We implement a VSS example on three edge systems for comparison, e.g., Raspberry Pi3, Raspberry Pi4, and NVIDIA Jetson Nano. The monitored power consumption of these systems is close to [1], [3]. These edge systems (Raspberry Pi3, Pi4, and NVIDIA Jetson Nano) represent typical constrained computing platforms widely used for deploying edge-AI applications, making them appropriate benchmarks to demonstrate ICE's advantage in realistic scenarios.

V. Experiment Results

We validate ICE using a combination of circuit simulations for modified NAND blocks, representative test-chip measurements for TLC sensing robustness, and system-level modeling and evaluation on edge platforms.

A. Implementation of nvIMC 3D NAND Techniques

To explicitly tackle the complexity issues in analog designs highlighted in Motivation (❶), ICE reuses existing NAND logic components to implement our digital page multiplier instead of numerous ADCs and DACs besides the NAND array. And explicitly targeting Motivation (❷), ICE proposes a newly added TC accumulator to achieve signed integer computations through bit inversion and bias addition, thus eliminating the need for complex subtractor circuitry.

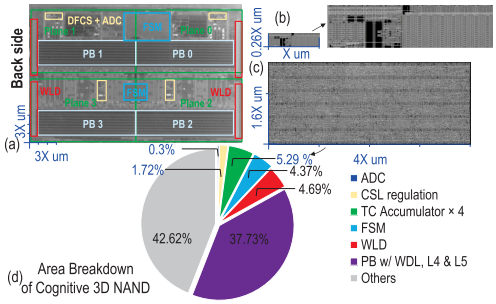


Fig. 14. (a) Backside die photo of the cognitive 3D NAND test chip used to validate representative nvIMC circuit components. (b) Layout of an 8-bit ADC. (c) Layout of a TC digital accumulator. (d) Area breakdown of modified and reused peripheral circuits.

TABLE II
Comparison of Prior Arts and This Work

	This Work	IEDM 2019 [18]		IEDM 2018 [16]	
Precision	INT8/INT4	UINT4	UINT8 (Re-de.)	1b	UINT8 (Re-de.)
Type	Digit	Current		Voltage	
Level of Cell	TLC	SLC		SLC	
Dimension #	512	512		512	
Total ADC #	0	N/A	512×4	N/A	512×16
Accumulator #	4	1		1	
Total VVM per run #	16/128	N/A	1	N/A	1
tANA (us)	N/A	1	50	2	50
tR_pSLC (us)	50 (4-plane)	N/A		N/A	
tMUL (us)	2 (4-plane 1b)	N/A		N/A	
tACC (us)	35 (4-plane 1b)	N/A	4	N/A	8
tAvg.VVM (us)	9.25/1.16 (4-plane parallel)	N/A	54	N/A	58
Output Precision	Full Precision	N/A	14b	N/A	15b
Per ADC (mW)	N/A	2.57(4b)	3.3(8b)	2.57(4b)	
WLD (mW)	33 (4-plane)	33 (4-plane)		33 (4-plane)	
Page Buffer & Array (mW)	112 (4-plane)	0.0059	0.149	0.0026	0.541
1b MUL in Page Buffer (mW)	12.4	N/A		N/A	
FSM (mW)	23.6	23.6		23.6	
ACC (mW)	26.4	N/A	8	N/A	8
GCR (mW)	N/A	N/A		11	
Others (mW)	49.5	49.5		49.5	
Avg. E/VVM (uJ)	6.17/0.77	N/A	371.1	N/A	1228.4
Additional Area Overhead (%)	15.4	4460		546	

*Note: Analog Computing (ANA), Multiplication (MUL), Accumulation (ACC), Global Current Regulator(GCR), Average (Avg.), Power (P), Energy (E), Re-design (Rede.). Results are performed at the TT corner, and 25 °C.

We implement a cognitive 3D NAND test chip using the CuA technology for verifying our proposed VVM operation shown in Fig. 14(a). Follow [14], each plane is placed one DFCS followed by an 8b ADC. The ADC structure can be referred to [20]. Fig. 14(b) shows the layout of the 8b ADC. Through measurement, we observe the ADC increases 0.03% the area overhead of the 3D NAND test chip. Fig. 14(c) demonstrates the one TC accumulator. The synthesized TC accumulator only increases by 1.6%. The area overhead of the modified page multiplier (with WDL, L4, and L5), and TC accumulator is 15.4% of the test chip. It proves that our proposed design is more feasible in commercial 3D NAND chips compared to other two prior works.

Table II compares prior analog 3D NAND nvIMC works [16], [18] with our fully digital solution. To ensure a fair and meaningful comparison, all designs are normalized to the same end-to-end task: a complete 512-dimensional signed

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VVM with INT4/INT8 precision, which is the fundamental kernel required by real-world vector similarity search (VSS) workloads. This normalization is critical, as several prior analog nvIMC works report latency and energy figures for partial or idealized analog operations, without accounting for the full sequence required to generate a complete VVM result.

In analog nvIMC architectures, VVM computation relies on current summation along the CSL or BL, followed by ADC quantization. To produce a single 512-dimensional VVM result, such designs must perform (i) read-like operations to initialize and stabilize analog cell or sensing currents (e.g., at biased V_{BL}), (ii) wide-range current regulation using CSL regulators, and (iii) digitization of partial sums using a large number of ADCs. When these mandatory stages are included, the effective latency, energy consumption, and area overhead increase substantially and must be considered in a task-complete comparison.

Under this unified workload definition, we re-evaluate the architectures in [18]. After redesigning the computation flow to support a full 512-dimensional UINT8 VVM with four-plane decomposition, the design requires 2,048 ADCs, where each 8-bit ADC quantizes a 15 × 15 cell-current summation corresponding to two unsigned unary 4-bit operands. We observe that deploying 2,048 ADCs alone increases the area overhead by 155.8% relative to a standard 3D NAND test chip. Moreover, the overhead of CSL current regulation prior to ADC quantization, which is not included in the original evaluation, dominates the peripheral cost. When CSL regulation with DFCS is incorporated, the required 2,048 CSL regulators contribute an additional 4,300% area overhead, resulting in a total area overhead of 4,458% for enabling in-memory computing. This analysis indicates that placing ADCs at the CSL side is not scalable for high-dimensional VVM.

For the BL-side ADC approach in [16], relocating ADCs reduces per-ADC complexity but requires a substantially larger number of ADCs to support full-precision computation. After redesigning the architecture to generate a complete 512-dimensional UINT8 VVM with four-plane decomposition, the design requires 8,192 ADCs, i.e., four times more than [18]. Each 4-bit ADC quantizes the result of a 1-bit input and 4-bit weight multiplication in a single NAND string, followed by digital accumulation of partial products. We further synthesize the required 128-bit digital adders to evaluate accumulation latency, which is approximately 4 μs for [18] and 8 μs for [16] under a 30 ns clock period.

In contrast, our digital approach eliminates ADCs, CSL current regulation, and analog summation entirely by performing latch-based computation inside the page buffer. As shown in Table II, the proposed cognitive 3D NAND meets Motivations 1 and 2 by significantly reducing area overhead, circuit complexity, and energy consumption. By retaining sensed data in the weight-data latch (WDL), ICE enables read–compute pipelining and BET-based VVM without re-initialization overhead. INT4 BET VVM requires 148 μs for four iterations of 1-bit AND and accumulation, while INT8 BET VVM achieves the same latency by processing two 4-bit groups in parallel. For INT8 precision, ICE is 60× more energy-efficient than [18] and 199× more energy-efficient than [16], while maintaining full 25-bit output precision (compared to 14-bit and 15-bit outputs, respectively). With minimal peripheral modifications, the proposed design is compatible with commercial TLC 3D NAND technology and is validated through a combination of representative silicon measurements, post-layout circuit simulations, and system-level modeling, demonstrating the feasibility of the proposed digital in-NAND VVM framework.

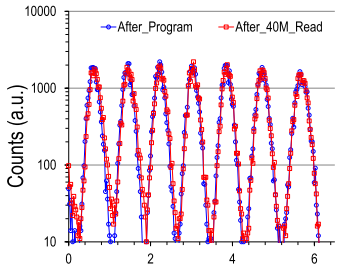


Fig. 15. Measured GC133 V_t window.

TABLE III
COMPARISON OF SIMULATED ACCURACY

	CelebA(20K)	
	Dimension #, Compute Format	Accuracy
von Neumann solution	512, L2 (FP32)	95.1%
	512, L2 (INT4)	95.1%
This work	512, ideal BET VVM (INT4)	95.1%
	512, BET VVM (INT4) w/ 0.5% RBER	95.054%
	512, BET VVM (INT4) w/ 0.7% RBER	95.114%
	512, BET VVM (INT4) w/ 1% RBER	95.129%
	VGGFace2(170K)	
	Dimension #, Compute type	Accuracy
von Neumann solution	512, L2 (FP32)	93.8%
	512, L2 (INT4)	93.7%
This work	512, ideal BET VVM (INT4)	92.6%
	512, BET VVM (INT4) w/ 0.5% RBER	92.098%
	512, BET VVM (INT4) w/ 0.7% RBER	91.352%
	512, BET VVM (INT4) w/ 1% RBER	91.036%
	CASIA-WebFace(500K)	
	Dimension #, Compute type	Accuracy
von Neumann solution	512, L2 (FP32)	84%
	512, L2 (INT4)	83.8%
This work	512, ideal BET VVM (INT4)	81.5%
	512, BET VVM (INT4) w/ 0.5% RBER	80.767%
	512, BET VVM (INT4) w/ 0.7% RBER	79.595%
	512, BET VVM (INT4) w/ 1% RBER	79.576%

B. Validation of Overall Accuracy of Using HTNS

To specifically address Motivation (❸), which emphasizes error tolerance without incurring ECC overhead, the introduced BET data encoding and BET VVM are designed to tolerate error bits in the absence of ECC support during VVM computation. We implement our 1-3-3 Gray code (GC133) in our nvIMC 3D NAND test chip using the CuA technology shown in Fig. 15. With GC133, we only deploy V_{READ3} to distinguish LV_t cells and HV_t cell, which is more reliable than 1-2-4 Gray code and faster than 2-3-2 Gray code, as shown in Fig. 15. It is hard to make a perfect pass during programming by adopting GC133 only. The programming sequence generally gives a program pass if the fail bits are lower than a setup fail-bit threshold. It means some error bits needed to be programmed into HV_t will stay in LV_t, which deteriorates the computing reliability. We conduct 40 million read operations to observe the influence of read disturbance. The experiment result shows that it only causes a slight V_t shift after 40 million times reads, which proves the storage feasibility.

Table III shows FP32 baseline accuracy for CelebA, VGGFace2, and CASIA-WebFace at 95.1%, 93.8%, and 84.0%. INT4 maintains similar accuracy (95.1%, 93.7%, 83.8%). With ideal BET VVM, accuracy drops slightly

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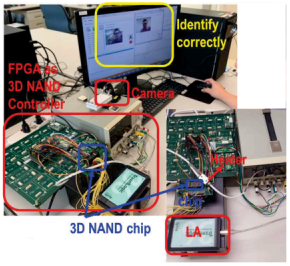


Fig. 16. A proof-of-concept system.

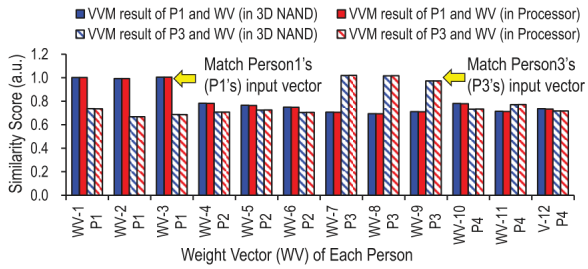


Fig. 17. Measured similarity scores produced by the cognitive 3D NAND test chip and by a processor-based implementation.

(0.05%, 1.1%, 2.3%). Across RBER = 0.5%, 0.7%, 1%, accuracy remains robust; CelebA even improves slightly due to BET redundancy and dataset traits. At 1% RBER, accuracy is 95.1% (CelebA), 92.1% (VGGFace2), and 80.8% (CASIA), showing minimal degradation.

We further built a proof-of-concept system, shown in Fig. 16 validates in-memory computing using a cognitive 3D NAND test chip. The similarity scores (i.e., VVM results) are measured by a tester that accesses the NAND device through the standard ONFI 5.0 interface. Specifically, the Field-Programmable Gate Array (FPGA) tester enters the NAND engineering mode and issues vendor-specific commands to trigger GC133 sensing, in-NAND VVM, and result accumulation. The computed similarity results are then read out through the conventional page-buffer read path, without any modification to the standard NAND I/O interface. Fig. 17 also shows the measured results. The x-axis is four persons' weight vectors, each with three weight vectors. The similarity scores using our solution are almost the same as the processor, and the most similar person has much higher scores than others. Thanks to the help of GC133, and BET scheme, leveraging nvIMC 3D NAND for scoring similarity could mitigate the ECC overhead even with a minor computing mismatch.

C. System Comparison

Fig. 18 and Fig. 19 compare the total execution time and energy consumption using three basic edge systems; each edge system contains an ICE with four cognitive 3D NAND chips, or an ICE with four chips using the previous work [18]. The proposed ICE with four nvIMC 3D NAND chips power Spec. is shown in Table IV. These results confirm ICE effectively minimizes unnecessary data movements and enhancing search efficiency. With the configuration of the heavyweight AI model (FaceNet) and the small dataset (CelebA), these edge devices using ICE with digital nvIMC techniques can enhance the execution time by 1.01×-1.4× and the energy efficiency by 1.02×-1.3×. With the configuration of the lightweight AI model (MobileFaceNet) and the CelebA dataset, using ICE can enhance the execution time by 1.04×-1.6× and the energy efficiency by 1.02×-6×. On the other hand, with the configuration of FaceNet and the VGGFace2 dataset, these edge devices using ICE can improve the execution time by 1.9×-8.1× and the energy efficiency by 1.7×-5.2×. With the configuration of MobileFaceNet and the VGGFace2 dataset, using ICE can enhance the execution time by 1.7×-13.9× and the energy efficiency by 1.4×-11.1×.

TABLE IV
ICE W/ THE COGNITIVE 3D NAND CHIP POWER

Edge Storage System Example	the Intelligent Cognition Engine (ICE) w/ 4× nvIMC 3D NAND chips	
Component	**Power**	**Unit**
2× Controllers	158	mW
128KB SRAM Buffer	105	mW
3D NAND Interface	80	mW
4× cognitive 3D NAND chips	853.3	mW
ICE Total Power (VSS)	1196.3	mW

Note: These parameters are from the datasheet report. SRAM Buffer is from CACTI.

On the contrary, work in [18] needs many ADCs to accomplish VVM operations. It leads to significant energy consumption and a large area overhead compared to our proposed solution shown in Fig. 19. With the configuration of MobileFaceNet and the CASIA-WebFace dataset, using the previous work [18] improves the execution time by 10.8×-56.6× and the energy efficiency by 6.2×-30.4×. With the same configuration, using ICE can improve the execution time by 17.8×-122.6× and the energy efficiency by 11.5×-162×. Considering the overall system performance, the speed (resp. energy efficiency) of ICE is higher than that of [18] by 1.16×-6.48× (resp. 1.24×-18.8×). Furthermore, our proposed cognitive 3D NAND also outperforms [18] in other evaluation metrics such as area (39.5× smaller) since [18] has to deploy numerous ADCs. To sum up, using the lightweight MobileFaceNet can decrease tFE, which may increase the percentage of tSR with a large dataset. Under large dataset scenarios (VGGFace2, CASIA-WebFace), ICE significantly reduces tSR and E_{SR} by avoiding frequent data transfers. ICE then outperforms previous works substantially due to its hierarchical search and digital nvIMC implementations. Although we use the INT8 format here, it is reasonable to infer that ICE can still outperform other works in different INT-based formats, such as INT4.

For other applications with large datasets, the data movement and swap deteriorate the overall performance. At the same time, DRAM capacity is limited, as shown in Fig. 18 and Fig. 19. This phenomenon and characteristic are significant for our target environment, commonly resource-constrained edge devices. To address these issues, ICE could serve in a computational storage unit to compute data locally without triggering additional data movement and swap. As discussed in Sec. III-B, the HTNS strategy reduces unnecessary data access effectively. Our experimental results confirm significant performance improvements, with reductions in search time up to 122.6× and improvements in energy efficiency by up to 162×. ICE can process the similarity search in parallel with computing units to improve overall performance, while the processors in edge systems can utilize DRAM completely for other complicated computations.

Unlike volatile IMC solutions (e.g., SRAM-based) that suffer from high leakage and limited density, or emerging non-volatile IMC (e.g., ReRAM) that faces challenges in multi-level analog precision and write endurance, ICE leverages the high-density TLC 3D NAND. By adopting a digital path, ICE avoids the massive area overhead of ADCs (>4000%

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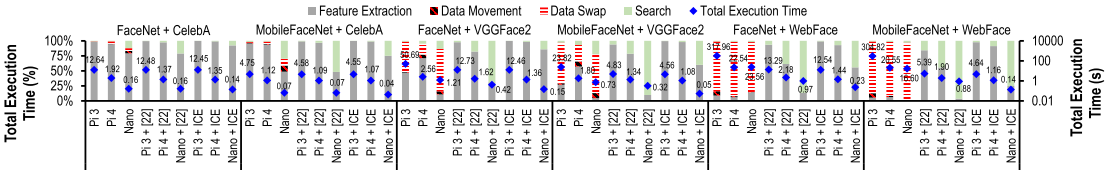


Fig. 18. Comparison of the measured total execution time of edge systems and simulated total execution time of systems with [18] and ICE.

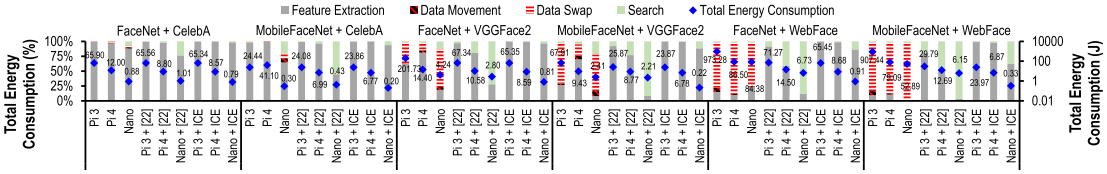


Fig. 19. Comparison of the measured total energy of edge systems and simulated total energy of systems with [18] and ICE.

in analog designs) and remains immune to the PVT sensitivity that plagues current-mode summation.

VI. Conclusion

This paper presents the Intelligent Cognition Engine (ICE), a low-energy and scalable cognitive 3D NAND framework for vector similarity search. The proposed digital nvIMC techniques extend conventional page-buffer operations and fail-bit counter-based two's-complement accumulation to enable signed vector computations while avoiding the complexity of ADC/DAC-based analog designs. By combining bit-error-tolerant encoding with digital in-NAND computation, ICE improves robustness to storage-induced errors without invoking ECC during online VVM execution. Based on circuit design analysis and post-layout evaluation, the proposed approach incurs an estimated area overhead of approximately 15% in a circuit-under-array (CuA) 3D NAND implementation to support INT8/INT4 VVM operations. Together with system-level orchestration, ICE enables similarity computations to be performed closer to data storage, thereby reducing data movement and mitigating frequent DRAM swapping on memory-constrained edge platforms. In addition, the hierarchical dataflow organization and Top-N search strategy help reduce unnecessary data accesses and accelerate the retrieval of the most relevant results. The effectiveness of ICE is evaluated through a combination of post-layout circuit simulations, representative silicon measurements, and system-level modeling on commercial edge devices, including Raspberry Pi 3/4 and NVIDIA Jetson Nano. Experimental results indicate that ICE can improve system execution time by 17.8× to 122.6× and energy efficiency by 11.5× to 162× compared to conventional von Neumann-based approaches. These results demonstrate the feasibility and potential of fully digital 3D NAND-based nvIMC architectures for accelerating vector similarity search in memory-constrained edge AI environments.

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Han-Wen Hu (Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from National Chung Cheng University, Taiwan, in 2014 and 2016, respectively. He is currently pursuing the Ph.D. degree in electrical engineering with National Tsing Hua University, Taiwan. In 2019, he joined the Emerging System Laboratory, a Division of Macronix International Company Ltd., where he focuses on 3D NAND circuit design, page-buffer-based computing architectures, and system-level optimization for next-generation non-volatile memory technologies. His work spans device-to-architecture co-design, analog-digital hybrid computing, and performance-energy optimization for large-scale data-processing applications. His research interests include memory architectures, in-flash and in-memory computing, approximate vector processing, and high-dimensional computing systems.



Yuan-Hao Chang (Fellow, IEEE) received the Ph.D. degree in computer science from National Taiwan University. He is currently a Professor with the Department of Computer Science, National Taiwan University. His research interests include computer systems, computer architecture, embedded systems, operating systems, memory systems, storage systems, and non-volatile memory.



Bo-Rong Lin received the M.S. degree from the Institute of Computer and Communication Engineering, National Cheng Kung University, Tainan, Taiwan, in 2019. He is currently the Project Manager with the Emerging System Laboratory, a Division of Macronix International Company Ltd., Hsinchu, Taiwan. His research interests include in- and near-memory computing, and system performance evaluation through simulation.



Huai-Mu Wang received the M.S. degree in electronics engineering from National Chung Cheng University, Chiayi, Taiwan, in September 2019. He is currently a Staff Engineer with the Emerging System Laboratory, a Division of Macronix International Company Ltd., Hsinchu, Taiwan.



Yung-Chun Lee received the M.S. degree in electronics engineering from National Taiwan University of Science and Technology, Taipei, Taiwan, in June 2010. He is currently the Product Manager with the Emerging System Laboratory, a Division of Macronix International Company Ltd., Hsinchu, Taiwan. His research interests include non-volatile memory, storage systems, and in- and near-memory computing.



Hsiang-Pang Li received the B.S. and M.S. degrees from Chung Yuan Christian University, Taiwan, in 1994 and 1996, respectively. He joined the Nonvolatile Memory Product Division, Macronix International Company Ltd., such as EPROM, mask-rom, and flash memory. He is currently an In Charge with the Emerging System Laboratory, Macronix International Company Ltd. His research interests include in- and near-memory computing, memory system architectures, embedded systems, and NVM applications.



Chung-Kuang Chen received the B.S. and M.S. degrees in electronics engineering from National Cheng Kung University, Tainan, Taiwan, in 1997 and 1999, respectively. He is currently the Product Manager with the Product Development Division, Macronix International Company Ltd., Hsinchu, Taiwan. His research interests include non-volatile memory circuits and system designs.



Tei-Wei Kuo (Fellow, IEEE) received the bachelor's degree in computer science from National Taiwan University, Taiwan, in 1986, and the Ph.D. degree in computer science from The University of Texas at Austin, Austin, TX, USA, in 1994. He was a Distinguished Professor in computer science and information engineering and an Interim President of National Taiwan University from 2017 to 2019. He is currently the CTO of Delta Electronics with the Department of Computer Science and Information Engineering, National Taiwan University, which is a global leader in power and thermal solutions. His research interests include flash-memory storage systems, non-volatile memory systems, and embedded systems. He is a fellow of ACM and the Chair of ACM SIGAPP. He received numerous recognitions, including the Academic Award of Taiwan Ministry of Education, the Humboldt Research Award, the Distinguished Research Award of Taiwan National Science and Technology Council, the Outstanding Technical Achievement and Leadership Award from IEEE TCRTS, and many best paper awards, including those from ACM/IEEE CODES 2019, 2022, and 2024. He is the founding Editor-in-Chief (EiC) of *ACM Transactions on Cyber-Physical Systems* from 2015 to 2021 and the General Chair of ESWEEK in 2025.



Meng-Fan Chang (Fellow, IEEE) received the M.S. degree from The Pennsylvania State University, State College, PA, USA, and the Ph.D. degree from National Chiao Tung University, Hsinchu, Taiwan.

Prior to 2006, he worked in the industry for over ten years. This included the design of memory compilers (Mentor Graphics; 1996–1997) and the design of embedded SRAM and Flash macros (Design Service Division of TSMC; 1997–2001). In 2001, he co-founded at IPLib, Taiwan, where he developed embedded SRAM and ROM compilers, flash macros, and flat-cell ROM products until 2006. He is currently a Distinguished Professor with National Tsing Hua University (NTHU), Hsinchu, and the Director of Corporate Research with Taiwan Semiconductor Manufacturing Company (TSMC), Hsinchu. His research interests include circuit design for volatile and nonvolatile memory, ultralow-voltage systems, 3-D-memory, circuit-device interactions, spintronic circuits, memristor logics for neuromorphic computing, and computing-in-memory for artificial intelligence. He was a recipient of several prestigious national-level awards in Taiwan, including the Outstanding Research Award from MOST-Taiwan, the Outstanding Electrical Engineering Professor Award, the Academia Sinica Junior Research Investigator Award, and the Ta-You Wu Memorial Award. He was a Distinguished Lecturer for the IEEE Solid-State Circuits Society (SSCS) and Circuits and Systems Society (CASS), as well as the Chair of the Nano-Giga Technical Committee of Circuits and Systems Society (CASS), and an Administrative Committee (AdCom) Member of the IEEE Nanotechnology Council. He has been serving as an Associate Editor for IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: REGULAR PAPERS, and IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS. He has also been serving on the Executive Committee for the International Electron Devices Meeting (IEDM), as well as the Subcommittee Chair for the International Solid-State Circuits Conference (ISSCC), the International Electron Devices Meeting (IEDM), DAC, the IEEE International Symposium on Circuits and Systems (ISCAS), the Very Large Scale Integration-Design, Automation and Test (VLSI-DAT), and Asia and South Pacific Design Automation Conference (ASP-DAC). He has been serving as the Program Director for the Micro-Electronics Program with the Ministry of Science and Technology in Taiwan, as well as the Chair of the IEEE Taipei Section, the Associate Executive Director for Taiwan's National Program of Intelligent Electronics (NPIE), and the NPIE Bridge Program.

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